with greater precision by subsequent observations. As illustrated in Figure 8.2b, if the balloon swells during some time interval, doubling in size for example, all spatial separations will double in size as well: pennies that were 1 inch apart will now be 2 inches apart, pennies that were 2 inches apart will now be 4 inches apart, pennies that were 3 inches apart will now be 6 inches apart, and so on. Thus, in any given time interval, the increase in separation between two pennies is proportional to the initial distance between them. And since a greater increase in separation during a given time interval means a greater speed, pennies that are farther away from one another separate more quickly. In essence, the farther away from each other two pennies are, the more of the balloon's surface there is between them, and so the faster they're pushed apart when it swells. Applying exactly the same reasoning to expanding space and the galaxies it contains, we get an explanation for Hubble's observations. The farther away two galaxies are, the more space there is between them, so the faster they're pushed away from one another as space swells.

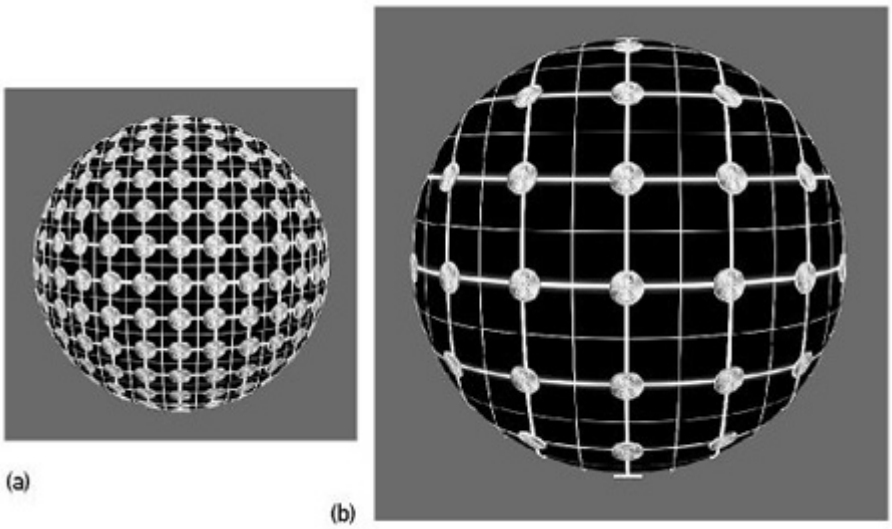


Figure 8.2 **(**a**)** If evenly spaced pennies are glued to the surface of a sphere, the view seen by any Lincoln is the same as that seen by any other. This aligns with the belief that the view from any galaxy in the universe, on average, is the same as that seen from any other. **(**b **)** If the sphere expands, the distances between all pennies increase. Moreover, the farther apart two pennies are in 8.2a, the greater the separation they experience from the expansion in 8.2b. This aligns well with measurements showing that the farther away from a given vantage point